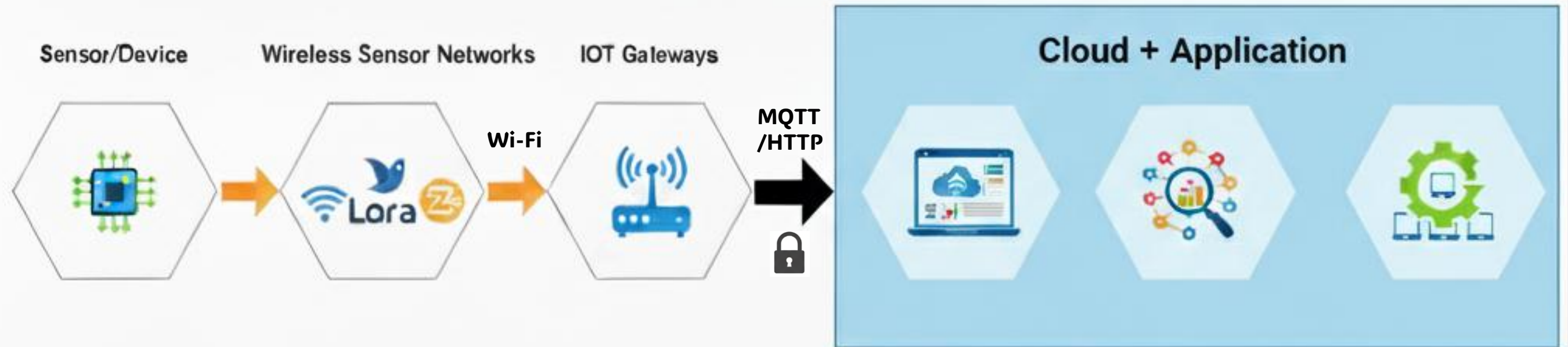
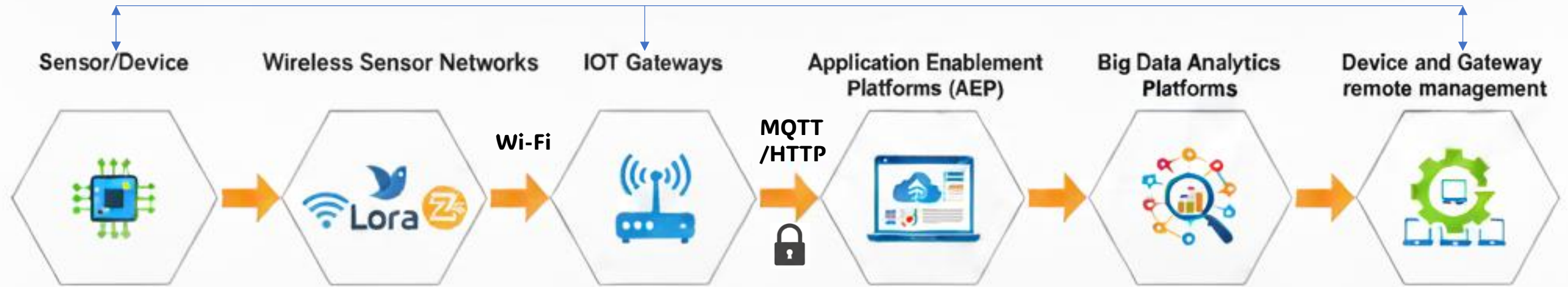




The Internet of Things: Connecting Our World

Session(5): IoT End-End Architecture

IoT End-to-End Architecture

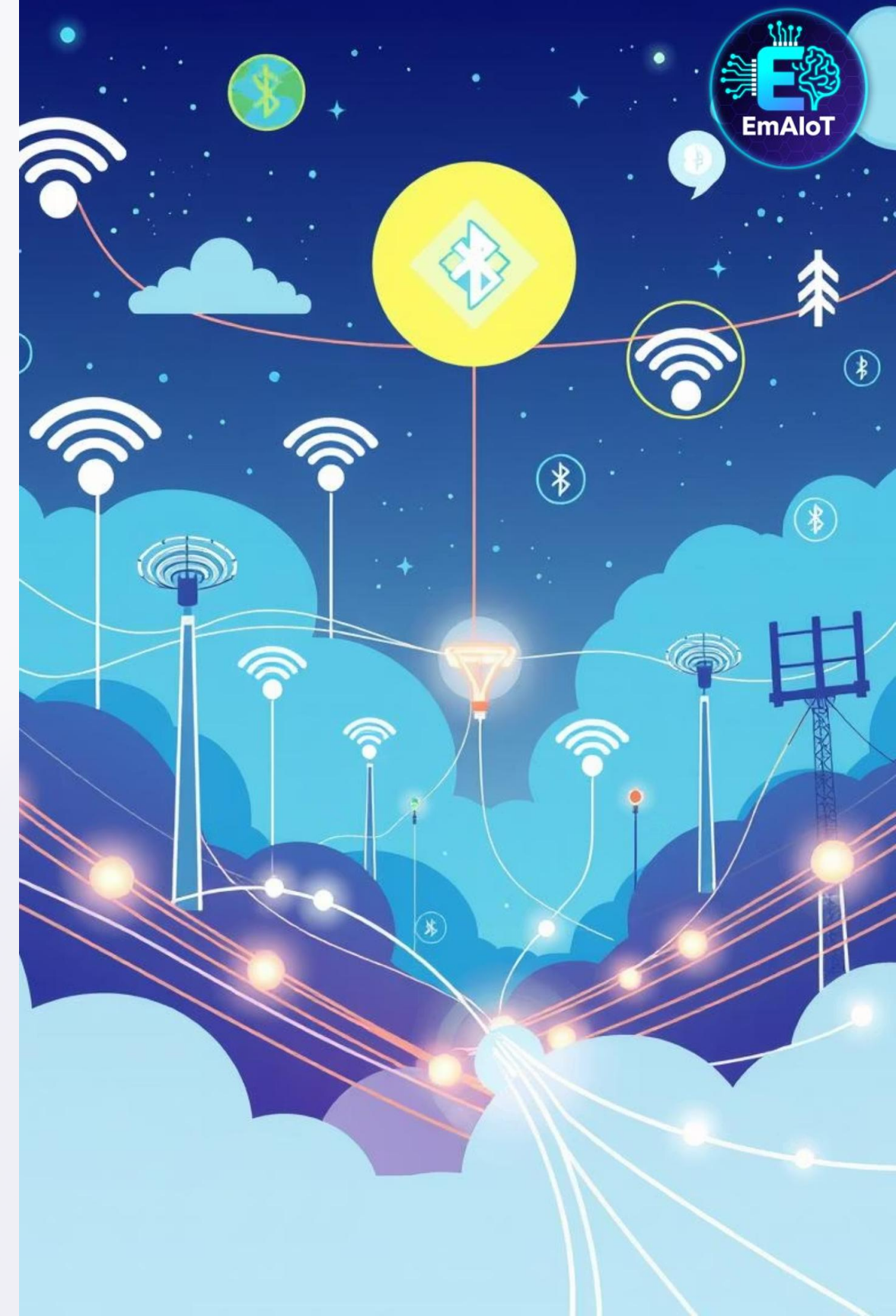


Devices and Sensors (Things)

- The starting point of IoT.
- Sensors and actuators measure the physical world: temperature, humidity, vibration, GPS, etc.
- Without devices, there is no data — and without data, there is no IoT.

Local Communication (Device → Gateway)

- Data needs to move from sensors to a nearby gateway.
- Technologies: **Bluetooth, Zigbee, Z-Wave, Wi-Fi, LoRa, wired protocols.**
- Choice depends on range, battery power, and data rate.



Gateway

- The hub that collects data from multiple devices.
- Functions:
 - Aggregation of data.
 - Protocol translation (different devices speaking different “languages”).
 - Preprocessing (filtering, compression, edge computing).
 - Temporary storage (if connectivity is down).
 - Security (encryption, authentication).
- Acts as the “local brain” of the IoT system.



Backhaul Communication (Gateway → Cloud)

- After preprocessing, data moves to the cloud via wide-area networks.
- Technologies: **Cellular (3G/4G/5G, NB-IoT), Ethernet, Wi-Fi, Satellite.**
- Protocols: **MQTT, CoAP, HTTPS.**
- Must balance cost, latency, and reliability.

Backhaul Communication (Gateway → Cloud)



Application Enablement Platform (AEP)

- The central layer that simplifies IoT development.
- Provides **APIs, SDKs, device registry, device twins**.
- Abstracts the complexity of devices and protocols.
- Enables developers to build applications without reinventing the wheel.

Big Data Analytics / AI

- Raw data becomes valuable information.
- Processing pipelines: ingestion, storage, stream processing.
- Machine Learning / AI applied for:
 - Predictive maintenance.
 - Anomaly detection.
 - Usage insights and forecasting.
- Can be real-time or batch analysis.



Device & Gateway Remote Management

- Essential for long-term sustainability.
- Functions:
 - **OTA/FOTA updates** (firmware and software).
 - Remote configuration and restart.
 - Monitoring device health, battery, and connectivity.
- Reduces field visits and operational costs.





Applications and Services

- The layer visible to the end-user.
- Forms:
 - Mobile apps, web dashboards, smart home interfaces.
 - Industrial dashboards (SCADA, ERP integration).
- Converts insights into **decisions, alerts, or automated actions**.



Cross-Cutting Concerns

- **Security & Privacy:** encryption, authentication, key management.
- **Reliability & Maintenance:** OTA plans, data buffering, resilient design.
- **Cost & ROI:** balance between hardware, communication, cloud services, and business value.

Ex: Smart Factory

- Sensor measures machine vibration.
- Sends data via **Zigbee** to a **Gateway**.
- Gateway filters and forwards data via **4G** to the **Cloud**.
- **AEP** registers the device and provides APIs.
- **Analytics** detects abnormal vibration and predicts machine failure.
- System triggers a **maintenance ticket** and sends an **alert** to the operator.
- At the same time, a **FOTA update** is pushed to the device.
- All visible on a **dashboard application**.

